

EXPERIMENT NO 8

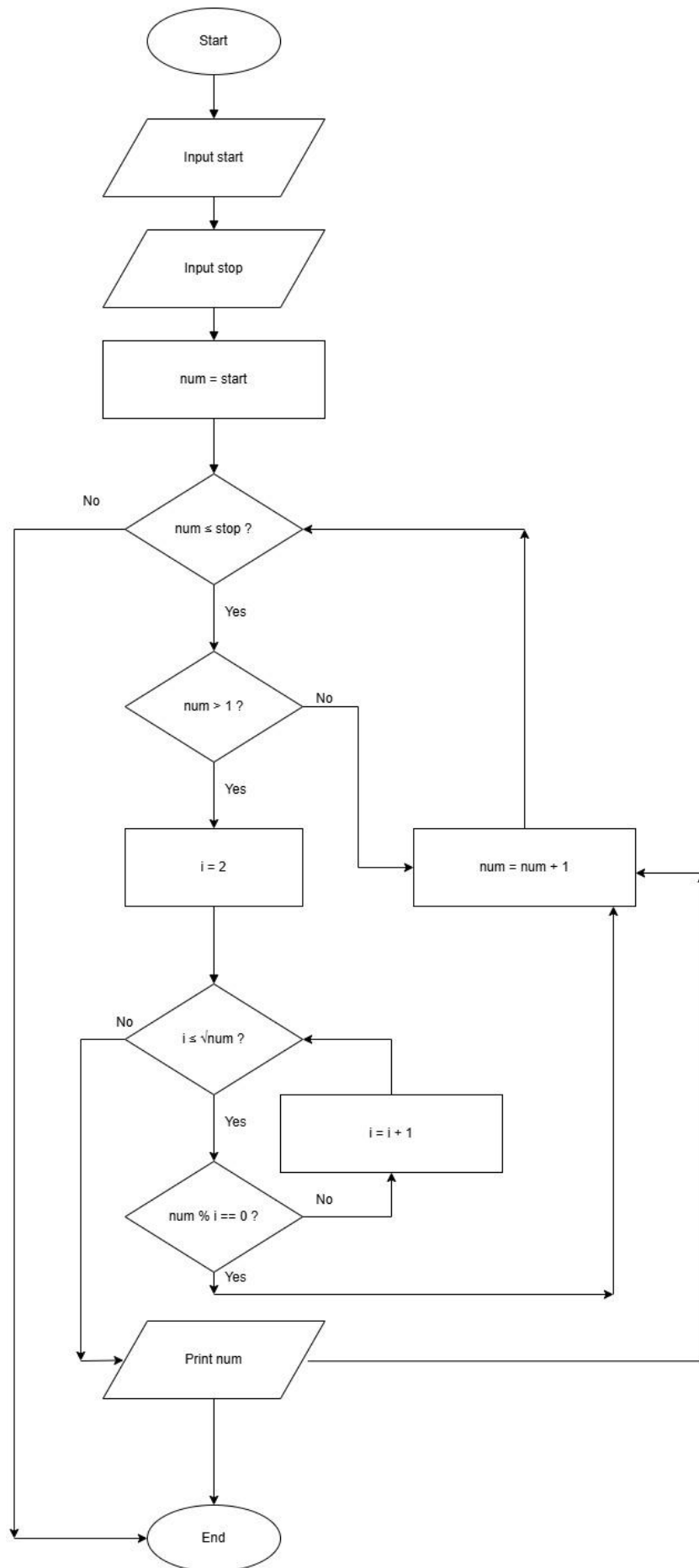
8.1.2 Prime Numbers in a Given Range

Algorithm

- Step 1: Start
- Step 2: Input start and stop values
- Step 3: Set f = False
- Step 4: For each number num from start to stop
- Step 5: If num > 1, assume it is prime (flag = True)
- Step 6: Check divisibility from 2 to $\sqrt{\text{num}}$
- Step 7: If divisible, set flag = False and stop checking
- Step 8: If flag is True, print the number and set f = True
- Step 9: After the loop, if f is False print "No primes"
- Step 10: Stop

Flowchart

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PYTHON CODE

```
start = int(input())
```

```
stop = int(input())
```

```
f = False
```

```
for num in range(start, stop + 1):
```

```
    if num > 1:
```

```
        flag = True
```

```
    for i in range(2, int(num ** 0.5) + 1):
```

```
        if num % i == 0:
```

```
            flag = False
```

```
            break
```

```
    if flag:
```

```
        print(num)
```

```
        f = True
```

```
if not f:
```

```
    print("No primes")
```

EXCECUTION

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8.1.2. Prime Numbers in a Given Range

Write a Python program to print all prime numbers within a given range (inclusive of both start and end value).

Input Format:

- First line contains an integer representing the start of the range.
- Second line contains an integer representing the end of the range.

Output Format:

- Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a new line.
- If no prime numbers exist in the range, print: "No primes".

Sample Test Cases